

1. Executive Summary

This simulation implements a privacy-preserving machine learning workflow using **Federated Learning (FL)** to train a model across four distributed healthcare clients (identified as P, L, H, and Br). The primary objective was to predict a binary clinical outcome (`outcome_predict1(2p-2mm_bin)`) without sharing raw patient data, sharing only model parameters (coefficients) with a central server.

The global model achieved an **accuracy of 74.44%** and a **Recall of 98.48%**, though the ROC AUC score of **0.6155** indicates moderate discriminative ability.

2. Methodology

- **Architecture:** The system uses the **Flower** framework to simulate a federated environment where 4 clients train local models on their private data partitions.
 - **Model Architecture:** A **Logistic Regression** model (Solver: `liblinear`) was selected for interpretability and binary classification suitability.
 - **Aggregation Strategy:** The server utilizes **FedAvg (Federated Averaging)** to aggregate local model parameters (coefficients and intercepts) into a global model after each training round.
 - **Privacy Mechanism:** Raw data (Excel files `client_P`, `client_L`, `client_H`, `client_Br`) remains on local client instances. Only the weight updates are transmitted to the server.
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3. Simulation Results

The simulation was executed over multiple rounds (likely 5, based on logs), concluding with an evaluation of the aggregated global model against a validation dataset of 90 samples.

Global Model Performance Metrics:

- **Accuracy:** 74.44%
- **F1-Score:** 0.8497 (Indicates strong performance on the positive class)
- **Precision:** 0.7471
- **Recall:** 98.48% (The model is highly sensitive, missing very few positive cases)
- **ROC AUC:** 0.6155
- **Weighted Loss:** 0.5604

Confusion Matrix Analysis: The model demonstrates a bias towards the positive class:

- **True Positives (TP):** 65
- **False Positives (FP):** 22 (High false alarm rate)
- **True Negatives (TN):** 2
- **False Negatives (FN):** 1

Analysis: While the model is excellent at identifying positive cases (high recall), it struggles to correctly identify negative cases (low specificity), misclassifying 22 out of 24 negative samples as positive.

4. Feature Importance Analysis

The notebook analyzes which clinical features drive the predictions by examining the logistic regression coefficients. Feature importance varies by client, reflecting local data distributions.

Client P (Top Features):

- **MachineName** (Coefficient: -1.1100): Strong negative impact on the outcome.
- **meanMLCSpeed** (-0.9919): Higher speed correlates with lower outcome probability.
- **GSmodulation** (-0.7905): Negative impact.
- **meanTGi** (+0.7250): Positive impact.

Client L (Top Features):

- **upperleaf** (-0.6033): Negative impact.
- **EdgeMetric** (-0.5721): Negative impact.
- **lowerleaf** (+0.5107): Positive impact.

This heterogeneity implies that different clinical factors may be more predictive depending on the specific institution (Client P vs. Client L).

5. Visualizations & Outputs

The notebook is designed to generate several visualizations to aid interpretation:

- **Precision vs. Recall Scatter Plot:** Compares performance across clients.
- **Loss Per Client:** Bar chart showing convergence.
- **Feature Importance Tables:** Rankings of top predictors.
- **ROC Curve:** Visualizing the trade-off between sensitivity and specificity.

6. Conclusion

The simulation successfully demonstrates the technical feasibility of Federated Learning for this healthcare dataset. While the system achieved high recall (98%), the high false positive rate suggests future work should focus on class balancing techniques or alternative model architectures to improve specificity and the ROC AUC score.